

Q1. a)

Domain: all $(x,y) \in \mathbb{R}^2$ with $x \neq 0$
 Range: all real numbers ($\mathbb{R}$) } 1

b)

Boundary: the line $x=0$ in $\mathbb{R}^2$
 Domain: open region
 (boundary not included!) } 1

c)

1 mark for correctly showing:

$$\lim_{(x,y) \rightarrow (0,0)} y^2 x = 0$$

possible Reasonings:

i) Epsilon-Delta Argument

Show that for any $\epsilon > 0$, if $\sqrt{x^2+y^2} < \delta$ then
 $|y^2 x| < \epsilon$ by approx. bounding the expression
 in terms of δ .

$$\text{For example; } |y^2 x| \leq |x| |y|^2 \leq \delta^3$$

So choosing $\delta < \epsilon^{1/3}$ works!

ii) Path wise approach:

$$\text{Along } x=0, \quad y^2 x = 0$$

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$$\text{Along } y=kx, \quad y^2 x = (kx)^2 x = k^2 x^3 \rightarrow 0 \quad x \rightarrow 0$$

Since all paths give limit at 0 the expression
 is continuous at 0, $\therefore$ the limit is 0.

iii) Using Polar Coordinates

$$x = r \cos \theta \quad y = r \sin \theta$$

Then we have

$$\lim_{r \rightarrow 0} r^2 \sin^2 \theta r \cos \theta$$

$$\Rightarrow \lim_{r \rightarrow 0} r^3 \sin^2 \theta \cos \theta \rightarrow \text{this is of the form } r \cdot g(\theta)$$

here as $r \rightarrow 0$

limit goes to 0.